

1. EXECUTIVE SUMMARY

THE GAP NOBODY FILLED

Sri Lanka produces thousands of university graduates every year — intelligent, ambitious and genuinely capable. Yet a significant proportion of them enter the job market with nothing to show beyond a transcript and a handful of generic academic assignments completed alone. No real projects. No collaborative experience. No proof that they can work with others to build something meaningful. Not because they lacked talent, but because nobody ever built them the infrastructure to do more than study.



WHAT COLLABHUB DOES

CollabHub is an AI-powered web platform built exclusively for university students in Sri Lanka to discover, verify and connect with the right collaboration partners for projects, hackathons, startups and skill building. It addresses a problem that is simultaneously a personal frustration, a cultural gap and a national economic issue.

HOW IT WORKS AND WHERE IT GOES

The platform does not simply connect students. It verifies what they can actually do through their real GitHub activity, matches them based on what their team is missing rather than what they already share, and creates a living record of collaboration that students can show to the world — to recruiters, to companies, to investors — as proof that they did not just study. They built.

CollabHub is being developed right now by a team of SLIIT undergraduates who are themselves the target users. It launches in Sri Lanka and is designed to expand across Southeast Asia as the first verified student collaboration ecosystem built specifically for emerging-market university talent.

AT A GLANCE

Category	Society, Work & Accessibility — Future of Work & Human Augmentation
Launch market	SLIIT, Sri Lanka — expanding to universities nationwide
Founding team	SLIIT undergraduates building for their own community
Long-term vision	First verified student collaboration ecosystem for emerging-market university talent

What follows sets out the problem in full, the product that solves it, the AI systems that make it trustworthy, and the realistic, staged path that takes CollabHub from a single-campus pilot to a regional platform.

2. PROBLEM STATEMENT

2.1 THE CULTURE THAT WAS NEVER BUILT

Walk into any Sri Lankan university today and you will find students who are sharp, curious and full of ideas. Ask them what they have built outside of their coursework and most of them will go quiet — not because they did not want to build anything, but because nobody ever showed them how to start, and because they could not find the right people to start with. The culture of building together simply does not exist here the way it does at universities in Singapore, India or the United States.

This is not an individual failure. It is a structural one. Sri Lankan university culture has been shaped almost entirely around assignments, examinations and GPA — the metrics of survival in an education system that was never designed to produce builders.

But within that system there is a quieter story that rarely gets told. There are students who feel the pull toward something more — the urge to build something real, to learn beyond the syllabus, to find people who want to grow and create and push themselves the way they want to push themselves. These students stay up reading about startups they wish they could build. They watch developers ship projects and wonder why they cannot seem to start their own. They have the ambition and the drive. What they do not have is direction, opportunity and, crucially, each other.

Because the hardest part of that feeling is not the lack of skills — skills can be learned. The hardest part is looking around a lecture hall of hundreds of students and not knowing who among them feels the same way. So they start alone: a new project, a few files, an initial commit, and then the questions begin. Nobody is going to see this. I don't know where this is going. Without another person beside them who shares the same hunger, the repository goes quiet, the idea fades, and motivation sustained entirely alone turns out to be one of the hardest things in the world to maintain. This happens quietly, repeatedly, across every university in Sri Lanka.

"That person who shares your hunger to build exists. They are at your university right now. CollabHub finds them for you."

CollabHub is built on a simple but powerful belief: that two students with the same hunger to improve, to build and to grow will achieve more together than either of them ever could alone. The right teammate is not just someone with a complementary skill set — they are someone who will message you at 11pm when you are about to give up, and remind you why you started.

2.2 WHAT HAPPENS WHEN STUDENTS GRADUATE WITHOUT PROOF

The Sri Lankan technology job market is facing a paradox. On one side are companies — local firms, startups, multinational offices — desperate for graduates who have collaborated on real projects, who understand version control under pressure, and who have shipped something, anything, into the real world. On the other side are graduates, many of them genuinely intelligent and motivated, who cannot demonstrate any of this. Their CV contains a GPA, a university name and a list of academic projects completed alone for a grade. No meaningful GitHub history. No evidence of initiative or collaboration.

The result is a job market that rewards the wrong signals. Recruiters, unable to verify real capability, fall back on university reputation and interview performance. Clever, ambitious students who never had the opportunity to build anything real get passed over for students who simply interview well. **This is not a recruitment problem — it is an infrastructure problem.**

2.3 EXISTING PLATFORMS DO NOT SOLVE THIS

Platforms that claim to address collaboration and talent discovery have existed globally for years. None of them were built for this user, this problem, this context:

Platform	Why It Fails
LinkedIn	Built for professionals documenting a career history. A first-year undergraduate with no work experience is invisible on it. Profiles are self-reported with no way to verify claims, and there is no feature for finding a project collaborator.
GitHub	Holds the most honest signal about a developer — real commit history — but it is raw, technical and inaccessible. Nobody browses GitHub profiles to find a hackathon teammate.
Devpost & similar	Exist only for specific events and disappear between them. No ongoing collaboration support, and no meaningful presence or adoption in Sri Lanka.
WhatsApp / Facebook	Where most Sri Lankan students actually find teammates today. Unstructured, unverified, limited to whoever is in the group, with no way to filter by skill or assess reliability.
Discord	Offers community but zero structure — no profiles, no verification, no intelligent matching, no persistent collaboration history.

The gap is not that these platforms are imperfect. The gap is that none of them were built for this specific user. CollabHub is built to fill that gap.

3. SOLUTION OVERVIEW

CollabHub solves the collaboration gap through three things working together as one unified system. These three elements remove the barrier to finding the right people, establish trust between strangers before collaboration begins, and create proof of initiative and teamwork that Sri Lankan graduates currently have no way to show.



Verified Proof of Work Profiles

- Connects to GitHub and reads real commit history
- Google Gemini generates a plain-language skill summary
- Students do not claim skills — their work history proves it

Complementary AI Matching

- Most platforms match similarity — CollabHub matches complementarity
- Surfaces people who have what your team is missing
- Mirrors how high-functioning real-world teams are actually built

Living Collaboration Record

- Every connection and project team formed becomes verifiable history
- Builds into a documented record visible to recruiters
- Proof of initiative and teamwork — not just academic grades

Together, these pillars remove the barrier to finding the right people, establish trust between strangers before a single line of code is written together, and generate the proof of initiative that Sri Lankan graduates currently have no way to show. A self-reported profile becomes a trustworthy one, and a lecture hall full of strangers becomes a searchable, verifiable talent network — one where the most useful teammate for your project is surfaced to you directly, instead of being left to chance.

1

Remove the barrier to finding the right people

2

Establish trust between strangers before collaboration begins

3

Create proof of initiative the job market can actually see

4. AI INNOVATION CORE

4.1 GITHUB VERIFIED PROFILE GENERATION

When a user connects their GitHub account through standard OAuth authentication — the same mechanism used by platforms like Devpost, Vercel and Netlify — CollabHub queries the GitHub GraphQL API to retrieve:

- Programming languages used across all public repositories
- Total commit count and commit frequency over the past six months
- Repositories created versus repositories with sustained multi-week activity
- Contribution calendar showing active days, streaks and gaps
- Repository descriptions and README content where available

This raw data is passed to the Gemini API through a structured backend prompt, which returns a clean, human-readable profile summary such as:

"Active Python and JavaScript developer. Consistent contributor over five months with 67 commits across four repositories. Two projects show sustained multi-week development suggesting strong follow-through behavior."

This summary appears on the student's profile alongside their academic information. Users who connect GitHub receive a **Verified Activity badge** and rank higher in discovery — transforming a self-reported profile into a trustworthy one automatically, with zero extra effort from the student.

4.2 RELIABILITY SIGNAL

Beyond raw skill detection, CollabHub surfaces a behavioral signal derived from GitHub activity patterns: does this person start repositories and sustain them, or do their projects go dark after the first two weeks? Are commits distributed consistently, or clustered into one frantic push before a deadline?

This matters because the most common reason student collaborations fail is not skill mismatch — it is **commitment mismatch**. A student who disappears three weeks into a project is a behavior problem that could have been anticipated if the right data was available. CollabHub makes that data visible before the collaboration begins, so students choose partners with open eyes rather than good intentions alone.

Reliability, made visible: sustained multi-week activity, streak consistency and follow-through are surfaced on every profile — the same qualities a hiring manager looks for, shown before the first message is ever sent.



4.3 COMPLEMENTARY SKILL MATCHING ENGINE

Each user profile contains a skill vector built from two sources: manually selected skill tags confirmed during onboarding, and languages and tools detected automatically from GitHub activity. When a student posts a collaboration request or browses for partners, CollabHub computes the gap between what the user has and what their project needs, then surfaces profiles that fill that gap most precisely.

This is not a machine learning model requiring months of training data — it is deliberate gap computation on structured profile data, technically achievable at the current stage of the platform and genuinely different in its logic from any matching system available to Sri Lankan students today.

4.4 AI MATCH EXPLANATION

When two profiles are identified as a strong complementary match, Gemini generates a brief plain-language explanation of why the match was made:

"You are building a web application and have indicated you need frontend support. This student has demonstrated consistent React activity across three repositories and is actively looking for a backend collaborator. Your skill sets are complementary and your availability windows overlap."



This removes the ambiguity from the decision to reach out. It makes the matching feel considered rather than random, and it gives both students a shared starting point for their first conversation — replacing the awkward cold message with a reason to talk that both people can already see.

Neither the matching engine nor the explanation layer requires months of model training: both run on structured profile data the platform already collects, which keeps the system fast to build, cheap to run and realistic for a two-person founding technical team to ship on the timeline this competition demands.

5. TARGET AUDIENCE

CollabHub is built for undergraduate students at Sri Lankan universities across all disciplines and all years of study. The immediate launch target is **SLIIT** — the university the founding team belongs to and knows most intimately — before expanding to University of Moratuwa, University of Colombo School of Computing, NSBM, IIT and other institutions nationwide.

Critically, CollabHub is not only for computer science students. A business student with a startup idea needs a technical co-founder. A design student needs a developer to bring their prototype to life. A data science student needs a domain expert to frame the right problem. The platform serves the full interdisciplinary spectrum, because the best teams — in hackathons, in startups, in the real world — are almost never built from a single discipline.

Secondary users are recruiters and companies who gain read access to a pool of students with verified, active profiles. Long-term geographic expansion targets Indonesia, Bangladesh, Vietnam and the Philippines — markets where the same collaboration gap exists at massive scale and no verified student collaboration platform currently operates.

6. PLATFORM FEATURES

6.1 CORE MODULES

Feature	Description
Authentication & Registration	Multi-step onboarding with GitHub OAuth to immediately populate verified profiles
Verified Student Profiles	GitHub-detected skills, interests, commitment level, reliability signal and Verified Activity badge
Live Collaboration Feed	Real-time feed of active requests; visitors see it read-only, creating urgency to register
Community Feed	Reddit-style discussion tab for questions, wins and feedback, with upvotes and comments
Browse & Discovery	Filter by skill, interest and commitment level; AI complementary matching surfaces the best profiles first
Connections & Chat	Private real-time messaging exclusively between mutually connected students, via Supabase Realtime
Notifications	Real-time alerts for expressed interest, accepted connections and new messages

6.2 THE RECRUITER LAYER

Students who opt in can make their CollabHub profile discoverable to verified company accounts. This gives recruiters access to something that does not exist anywhere else in Sri Lanka: a pool of students with verified GitHub activity, documented collaboration history and demonstrated initiative outside of academic requirements. For a recruiter tired of interviewing graduates with identical generic CVs, this is a meaningfully different signal — and it is entirely opt-in, so students stay in control of who sees their profile and when.

7. TECHNICAL APPROACH

7.1 TECHNOLOGY STACK

Layer	Technology
Frontend	React, Vite, Tailwind CSS
Routing	React Router
Data Fetching	TanStack Query
Forms & Validation	React Hook Form + Zod
Backend	Python, FastAPI, Pydantic
Database	PostgreSQL via Supabase
Auth + Realtime + Storage	Supabase
AI Processing	Google Gemini API
GitHub Data	GitHub GraphQL API
Frontend / Backend Hosting	Vercel / Render
Version Control	GitHub, feature branches merged via pull request

Every layer of this stack is a mainstream, well-documented, free-tier-friendly technology — chosen deliberately so a small student team can move fast without fighting its own tooling.



7.2 ARCHITECTURE & DEVELOPMENT METHODOLOGY

The Gemini API is called exclusively from the backend and never exposed to the browser, protecting credentials, enabling rate limiting and keeping all AI logic centralized and auditable. The platform is built API-first, so a future React Native mobile app will be served by the same backend without modification — one backend, web and mobile.

Development follows a deliberate, database-first, feature-by-feature build order — auth and schema, then profiles and GitHub OAuth, then collaboration and community feeds, then connections and chat, then AI matching, then notifications and polish — with each stage fully functional before the next begins, so the team always has a working, demonstrable core.

7.3 COST AND SUSTAINABILITY

CollabHub runs on free infrastructure tiers throughout development and early growth. At 1,000 active users, estimated monthly infrastructure cost is between **\$8 and \$35**. At 5,000 users, this remains between **\$33 and \$100** per month. The platform reaches thousands of real users before it costs anything meaningful to operate.

8. SOCIAL IMPACT

8.1 DEMOCRATIZING OPPORTUNITY

The collaboration gap is not equally distributed. Students from well-connected circles find teammates through existing networks; equally talented students from smaller institutions or non-technical disciplines are quietly excluded. CollabHub replaces network privilege with verified merit — the student with 200 consistent GitHub commits who nobody knows gets surfaced to the right collaborator as readily as the student who already knows everyone.

8.2 BUILDING THE GENERATION SRI LANKA NEEDS

Sri Lanka's technology sector cannot grow on graduates who have only completed academic assignments alone. CollabHub accelerates the formation of a generation of builders — people who leave university with a portfolio of real collaboration and the habits of initiative and teamwork the job market desperately needs.

8.3 A BRIDGE BETWEEN STUDENTS AND THE REAL WORLD

Every collaboration that happens on CollabHub is documented. Every team formed, every project posted, every connection made leaves a trace. Over time a student's CollabHub profile becomes a living record of what kind of builder and collaborator they actually are — visible to recruiters, companies, potential co-founders and investors. This changes the answer to the question every Sri Lankan graduate dreads in a job interview: *"What have you built outside of your studies?"*



9. PATH TO MARKET

9.1 GROWTH ROADMAP

Phase	Users	Focus
Phase 1	0 – 500	SLIIT only. Zero monetization. Prove retention and real usage.
Phase 2	500 – 2,000	Expand to Moratuwa, NSBM, Colombo. Sponsored hackathon listings and bootcamp partnerships begin.
Phase 3	2,000 – 5,000	Company talent-pool subscriptions. First meaningful revenue. Platform sustainability achieved.
Phase 4	5,000+	University partnerships, corporate hackathon hosting, regional expansion to Southeast Asia.

9.2 BUSINESS MODEL

CollabHub is permanently free for students. Revenue comes entirely from the institutional side: companies paying to access a verified talent pool, bootcamps advertising to a relevant audience, universities integrating the platform into career programs, and event organizers reaching an engaged student community. Nobody pays to find a collaborator — everyone benefits from a platform where finding the right one is simply easier.

10. CONCLUSION

There is a generation of Sri Lankan university students who are smarter, more ambitious and more capable than the opportunities currently available to them. They sit in lecture halls full of ideas they cannot act on because they cannot find the right people to build with. They graduate with transcripts that tell recruiters nothing meaningful about who they actually are. CollabHub exists because this does not have to be the story.

The platform is technically grounded and buildable with the current stack. The problem it solves is real and lived daily by its own founding team. The innovation — verified proof of work, complementary AI matching, a living collaboration record — is meaningful and does not exist anywhere in Sri Lanka today. The path to market is realistic, staged and financially sustainable without external funding through the critical early growth phase.

It means the best collaborator for your project is no longer whoever happens to sit next to you in lecture. It means a student from a small town with no connections but 200 honest GitHub commits gets found by the right person. It means Sri Lankan graduates walk into job interviews with something real to show.

It means the builders this country needs are no longer left behind because nobody built them the infrastructure to find each other. That is what CollabHub is for.